Null-space Control

When a robot has more degrees of freedom ($x \in \mathbb{R}^N$) than controllable output variables $(\text{e.g. } x \in R^6)$, the additional DOF can achieve secondary goals.

- e.g., Improving the controllability of the system (i.e. how far away the robot is from a singularity or joint limits)
- e.g., maximizing clearance to obstacles

Nullspaces are controllable subspaces that do not affect the output variables.

- Since position is output for a robot, we are saying that nullspaces are controllable actions of the robot that do not affect its output position.

Identifying the Nullspace of a Jacobian from SVD:

$$[\mathrm{U}, \Sigma, V^{\mathsf{T}}] = SVD\big(J(q)\big)$$

$$J \in \mathbb{R}^{m \times n}, \qquad U \in \mathbb{R}^{m \times m}, \qquad \Sigma \in \mathbb{R}^{m \times n}, \qquad V \in \mathbb{R}^{n \times n}$$

- For fat J matrix ($m < n$):

$$\Sigma = [diag(\sigma_1 \dots \sigma_m) \quad \mathbf{0}_{m-n}]$$

$$null(J) = span\{v_{m+1} \dots v_n\}$$

where v represents the columns of V

- If the Jacobian is singular, then at least one σ will be zero. In that case, include those eigenvectors into nullspace, i.e.,

$null(J) = span\{v_k \dots v_n\}$ for $\sigma_k = 0$, $k < m$

Projecting Control Objectives into the manipulator nullspace:

- Let $J^{\dagger}$ be the pseudoinverse (from SVD, or otherwise) for the linearized system
- Let $\dot{q}_2 = -\lambda \nabla_q U(q)$ be the differential feedback control signal found for *any* secondary tracking objective $U(q)$ to be minimized. λ is a scaling term. Then,

$$\dot{q}_{null} = \left(I_{N\times N} - J^{\dagger}J\right)\dot{q}_2$$

projects that behavior $\dot{q}_2$ into the nullspace as $\dot{q}_{null}$ such that

$$\dot{x} = J\dot{q}_{null} = 0$$

$\dot{q}_{null}$ can now be added to achieve some secondary control objective $U(q)$.

- i.e. avoid joint limits $\quad U(q) = (q_{\max} - q)^2 + (q - q_{\min})^2$
- …

Add up the different inputs:

$$\dot{q}_{total} = \dot{q}_1 + \dot{q}_{null}$$